BCS-082	Computational Complexity	L T P 3 0 0	3 Credits

Pre-requisites: students should know about that how to calculate the complexity of the programs.

Course Objectives:

CO1	To introduce the fundamentals of computational complexity theory.
CO2	To discuss basic concepts such as computational models, computational complexity measures (e.g., time and space complexity measures), complexity classes, reducibility and completeness notions.
CO3	To familiarize the concepts of randomized and approximation algorithms and discuss the related complexity classes
CO4	Understand Communication complexity
CO5	Understand Counting problems

Detailed Syllabus

MODULE-I

Models of Computation, resources (time and space), algorithms, computability, complexity. Complexityclasses, P/NP/PSPACE, reduction s, hardness, completeness, hierarchy, relationships between complexity classes.

MODULE-II

Randomized computation and complexity; Logical characterizations, incompleteness; Approximability. Circuit complexity, lower bounds; Parallel computation and complexity; Counting problems; Interactiveproofs.

MODULE-III

Probabilistically checkable proofs; Communication complexity; Quantum computation

Text Books:

1 Christos H. Papadimitriou., Combinatorial Optimization: Algorithms and Complexity Prentice-Hall

Reference Books:

1. SanjeevArora and Boaz Barak , Complexity Theory: A Modern Approach, Cambridge University Press.
2. Steven Homer, Alan L. Selman, Computability and Complexity Theory, Springer.

Course Outcomes: After the completion of the course the student will be able to:

CO1	The students will able to determine whether a problem is computable, and prove that some problems are not computable.
CO2	The students will able to categorize problems into appropriate complexity classes
CO3	The students will able to classify problems based on their computational complexity using reductions
CO4	The students will able to analyze optimization problems using the concept of interactive proofs
CO5	The students will able to analyze Counting problems